

SEAC 2026 Circular Statistics Workshop - Start Here

Sunday, 16:00-20:00

Instructor: Zlatko J. Kovačić

1. Open the RStudio project

Double-click or open in RStudio:

```
SEAC2026_Circular_Statistics.Rproj
```

Do not use `setwd()` and do not move individual files out of the workshop folder during the session.

2. Open the participant guide

```
01_WORKSHOP/SEAC2026_Workshop_Guide.pdf
```

3. Core exercise order

1. `01_Dataset1_Circular_Statistics_BEGINNER.R`
2. `02_Dataset2_Curvigram_BEGINNER.R`
3. `03_Dataset3_Ethno_Comparison_BEGINNER.R` - optional if time

Run code with:

```
Ctrl + Enter
```

Look for the labels:

```
RUN THIS  
CHANGE THIS  
INTERPRET THIS
```

4. Data

```
02_DATA/01_orientation_basics.csv  
02_DATA/02_megalithic_curvigram.csv  
02_DATA/03_ethno_directions.csv
```

All three datasets are synthetic teaching data.

5. Reference sheets

```
04_REFERENCE/Circular_Statistics_Cheat_Sheet.pdf
04_REFERENCE/Which_Method_Should_I_Use.pdf
04_REFERENCE/Curvigram_Explained.pdf
04_REFERENCE/AI_R_Assistant_Prompt_Card.pdf
04_REFERENCE/Further_Reading.pdf

04_REFERENCE/BOOK/Tadic&Kovacic~Orientation.pdf
04_REFERENCE/POSTERS/Circular_Statistics_Cheat_Sheet_Poster.png
04_REFERENCE/POSTERS/SEAC2026_circular.pdf
04_REFERENCE/POSTERS/SEAC2026_method.pdf
04_REFERENCE/POSTERS/Which_Method_Should_I_Use_Poster.png
04_REFERENCE/PRESENTATION/SEAC2026_Circular_Statistics_Workshop_Slides.pptx
```

6. Your exported results

Scripts save outputs under:

```
05_OUTPUT/Dataset1/
05_OUTPUT/Dataset2/
05_OUTPUT/Dataset3/
```

7. Two rules for the afternoon

Plot first, test second.

Do not interpret a statistical concentration as proof of intentional astronomical targeting.